

Knowledge Transfer: Agentforce Metadata Orchestrator

System Architecture & Implementation Guide

1. Executive Summary

This document details the architecture, implementation, and maintenance procedures for the headless Agentforce agent designed to automate the creation of Email-to-Case routing addresses.

Due to strict platform limitations (Agentforce operates strictly within the runtime data layer, restricting access to Setup Metadata), a custom architecture was engineered. This solution bridges the gap between the Agentforce runtime and the Metadata API using an External Client App (ECA), direct OAuth 2.0 token generation, and a patched legacy SOAP Apex wrapper.

2. System Architecture & The "Platform Wall"

The Core Limitation

Standard out-of-the-box Agentforce actions cannot read or write Setup Metadata. Furthermore, `UserInfo.getSessionId()` is systematically stripped of API permissions within the headless agent execution context, causing traditional authenticated Metadata API callouts to fail with `SESSION_ID_REMOVED` or `UNAUTHORIZED` exceptions.

The Custom OAuth Bridge Solution

To bypass the runtime context limitations, the system establishes a server-to-server OAuth flow to authenticate the Salesforce org back to itself:

- **External Client App (ECA) & Auth Provider:** Configured to manage a secure OAuth flow.
- **Direct Apex Callout:** Instead of relying on Named Credentials (which cannot inject tokens into XML SOAP bodies), Apex makes a direct REST callout to the ECA's `oauth/token` endpoint.
- **Remote Site Settings:** The org's own OAuth URL is whitelisted to permit this self-referential callout.

3. Component Implementation Details

A. The Access Token Fetch Mechanism

The Named Credential approach was attempted but ultimately abandoned because the Metadata API is a SOAP endpoint, requiring the authentication token to be placed within the `<SessionHeader>` XML tag, not just the HTTP Header.

Workflow: Apex executes an HTTP POST to `/services/oauth2/token`. The resulting Access Token is functionally identical to a UI Session ID and is manually injected into the SOAP request body.

B. Patching the Certinia MetadataService Class

The open-source `MetadataService.cls` was required to construct the SOAP XML. However, the legacy class (v42.0) could not parse modern metadata tags associated with Case Settings.

Manual Patch Applied:

- Upgraded the target endpoint to `/services/Soap/m/62.0`.
- Added missing metadata tags (e.g., `caseAutoProcUser`) to the Apex properties.
- Meticulously updated the `type_info` and `field_order_type_info` arrays to prevent runtime parsing exceptions when receiving modern SOAP responses.

C. Headless Invocation (Agent API)

To fulfill headless requirements, a dedicated **Connected App** was established specifically for the Agent API. This allows external systems to securely trigger the agent via OAuth 2.0 without human interaction.

4. Deployment & Maintenance Checklist

If deploying this solution to a new environment (sandbox or production), the order of operations is critical. Deploying out of order will result in authentication failures.

Step	Component	Action Required
1	Remote Site Settings	Whitelist the org's base URL and OAuth endpoint.
2	ECA & Auth Provider	Deploy the External Client App and configure the Auth Provider. Update callback URLs.
3	Apex Callout Class	Deploy the token-fetching logic that hits <code>oauth/token</code> .

4	MetadataService (v62)	Deploy the patched SOAP wrapper classes.
5	Agentforce Custom Actions	Deploy the agent orchestration logic, mapping I/O precisely to enforce deterministic execution.

5. Troubleshooting

- **LWC Errors in Agent Builder:** The Agent Builder UI can become unstable during Live Tests. Utilize the Salesforce CLI and the Agentforce Vibes VS Code extension for reliable compilation and deployment.
- **Agent Hallucinations:** If the agent pauses to prompt for input instead of silently executing, ensure the AgentScript syntax strictly enforces sequential steps (Retrieve → Generate Payload → Deploy) without interactive conversational loops.